

Meeting – July 2

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This Week

- Link Quality Analysis
- MeshMonitor

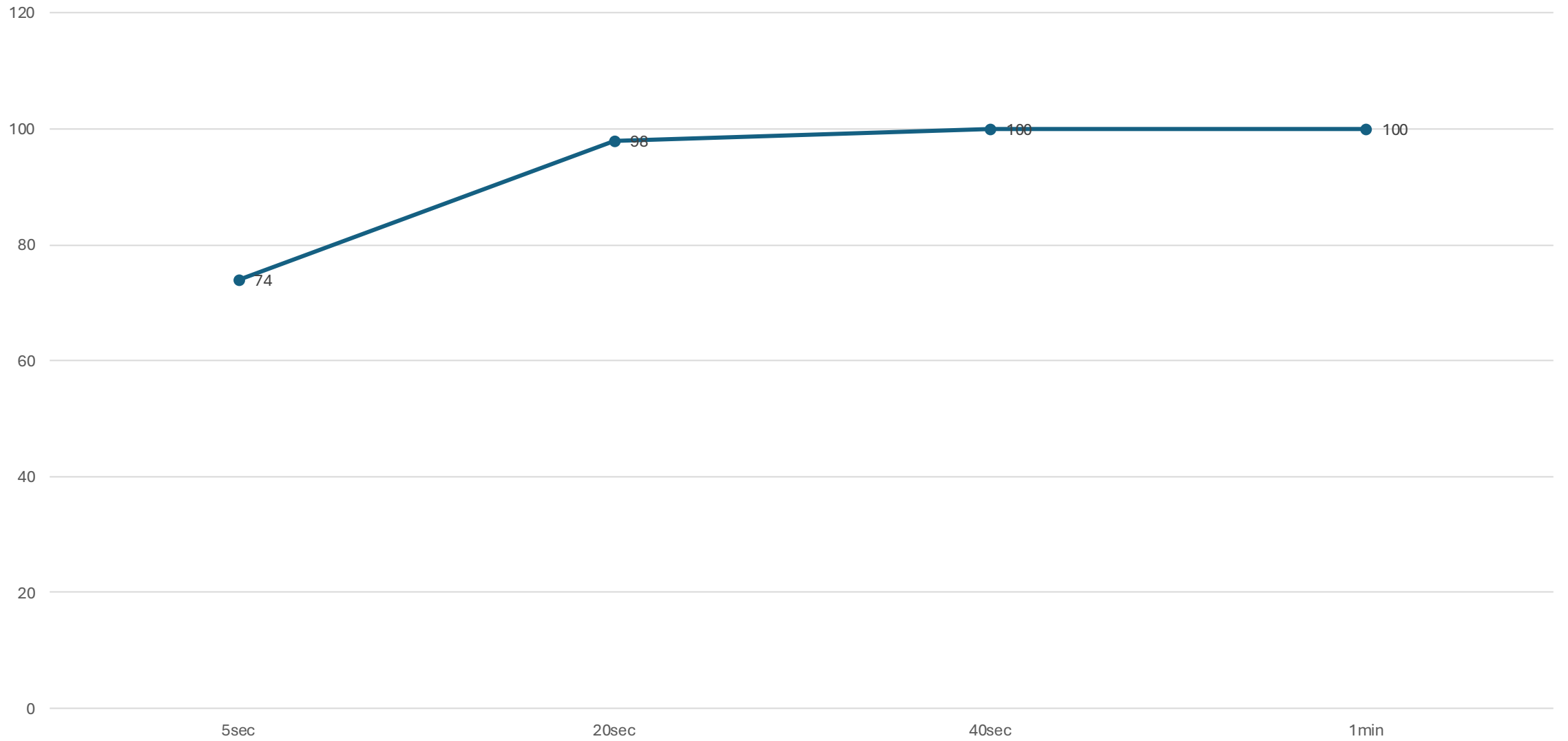
Link Quality Analysis

- This week, I continued my link quality analysis testing, gathering more data on how the link quality decreases as packets are sent at a higher interval.
- For each interval, 50 packets are sent between the Dalplex and the Goldberg. The intervals at which I am sending packets are: 5 seconds, 20 seconds, 40 seconds, 1 minute and 5 minutes.
- A test for each interval is run at three times of day: morning, afternoon and night.
- Although most of the tests are complete, not all tests for every time of day are complete yet. Because I had some initial issues with my scripts, the tests are taking longer than anticipated to complete.

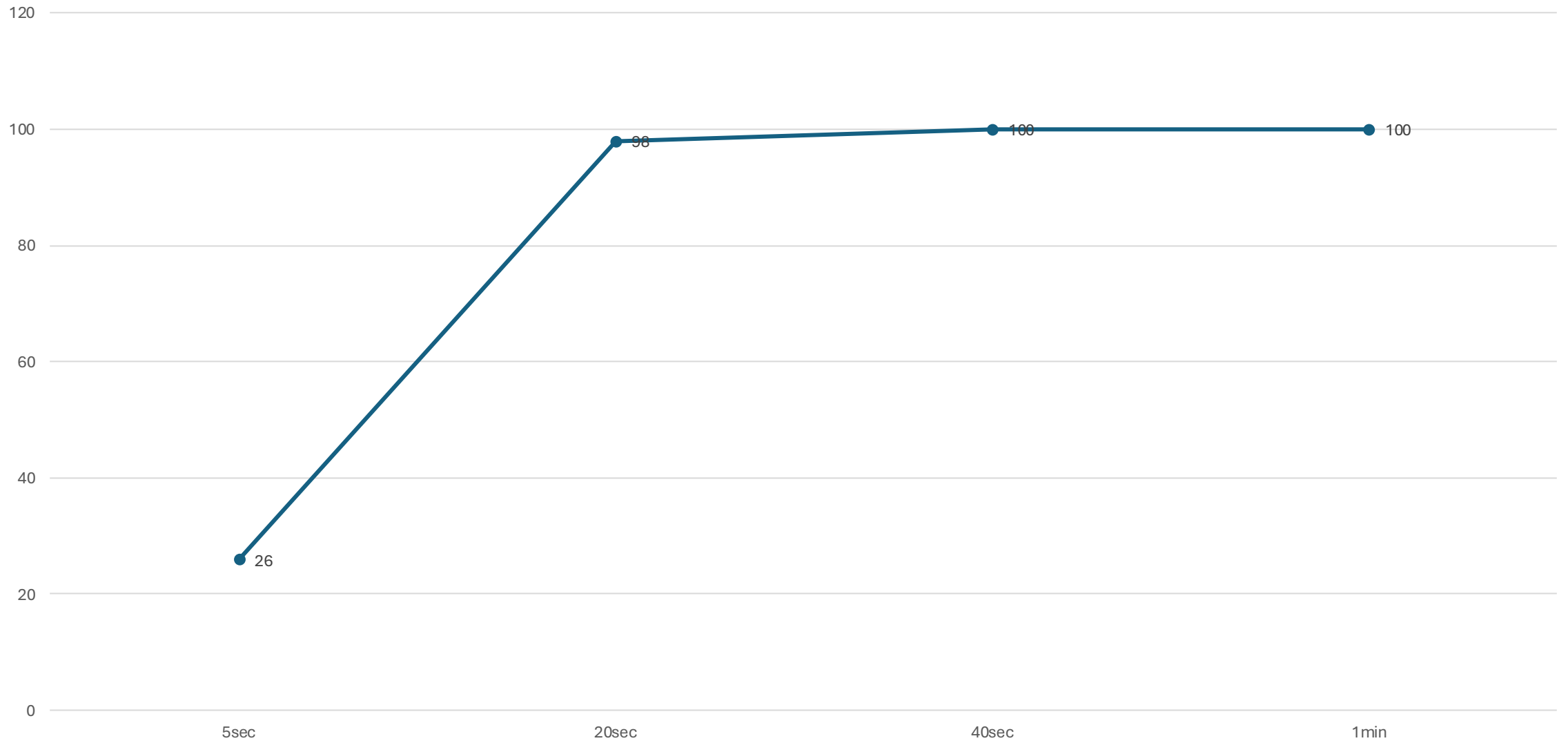
From Data Collected This Morning

METRIC/INTERVAL	5sec	20sec	40sec	1min	average
DR (%)	74	98	100	100	N?A
latency	N/A	N/A	N/A	N/A	N/A
avg hops	1.97297297	1.95918367	2	1.96	1.97303916
avg RSSI (outgoing)	-137.64865	-138.77551	-138.84	-139.12	-138.59604
RSSI SD (outgoing)	1.90345032	1.77089086	1.3606721	5.50191061	2.63423097
avg SNR (outgoing)	-12.797297	-13.897959	-14.09	-14.57	-13.838814
SNR SD (outgoing)	1.74487129	1.51804971	1.10744236	3.74371582	2.02851979
ACKs received	13	49	50	50	N/A
ACKs received (%)	26	98	100	100	N?A
avg RSSI (ACK)	-119.3	-121.35294	-124.62	-122.52083	-121.94844
RSSI SD (ACK)	13.0558629	11.1728663	14.5151434	15.1825622	13.4816087
avg SNR (ACK)	1.675	0.44607843	-2.45	-1.1197917	-0.3621783
SNR SD (ACK)	9.05691878	9.98890806	12.9188006	12.50319	11.1169543

DR (%)



ACKs received (%)



Morning Data cont.

- From last week's testing:
 - Delivery rate for both the 5 second and 1-minute intervals are up 48% from last week
 - ACKs received for the 5 second interval are identical to last week but are up 9% for the 1-minute interval.
 - The delivery rate for the receiver node at the Dalplex has increased drastically, while the number of ACKs being received by the sender node in the Goldberg is relatively similar.
 - What would cause this? Potentially a previous issue with the script now solved?

MeshMonitor

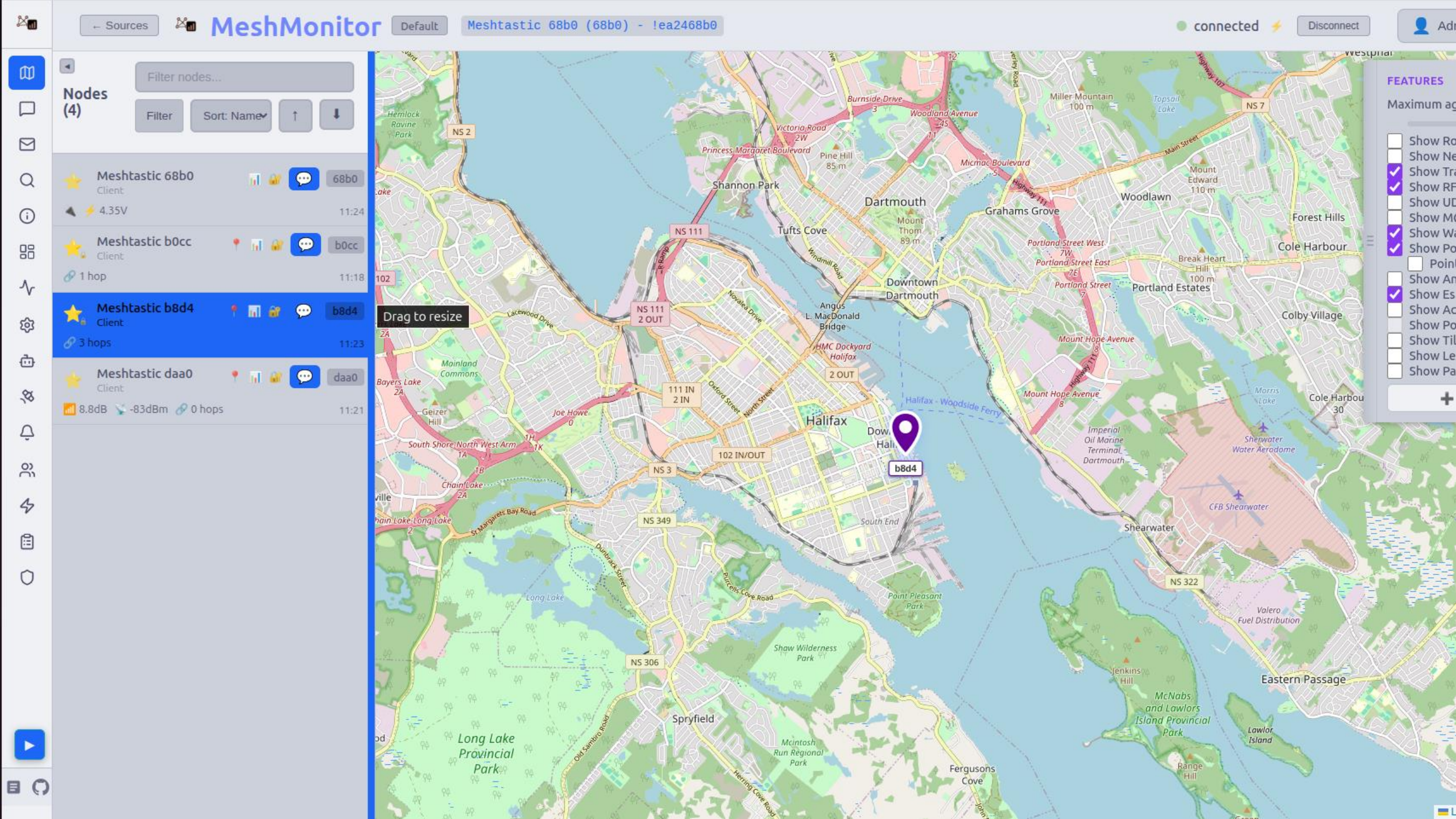
- While running the link quality tests, I focused my attention on other items, most notably, I focused my attention on implementing a community tool called MeshMonitor into the campus mesh.
- MeshMonitor is a web app used for the remote monitoring and administration of various kinds of LoRa mesh technology, including both Meshtastic and MeshCore.
- Currently, I have MeshMonitor working on my lab desktop, attached to a smaller LILYGO LoRa32 device, which has access to the mesh through the T-Beam set up in Goldberg Room 211.

MeshMonitor cont.

- MeshMonitor is a fantastic, highly useful tool, with a variety of features that significantly improve quality of life.
- MeshMonitor has a dashboard that visualizes absolutely every metric provided through local telemetry. This includes:
 - Packets TX/RX/TX relayed
 - Nodes by hardware model
 - Radio statistics (% of good, bad, duplicate packets)
 - Voltage + Battery Life
 - *And more!*
- There is also a dedicated telemetry dashboard which shows:
 - Node Status
 - Hop and distance distribution/heatmap across all nodes in the DB, color coded by SNR

MeshMonitor cont.

- MeshMonitor has a dedicated page for administrative commands, allowing the user to remotely change the settings of any device they have access to without using the terminal.
- Additionally, MeshMonitor has a security page, where MeshMonitor will scan the node DB for known low-entropy or duplicate keys
 - The issues causing low-entropy and duplicate keys have been patched.
- MeshMonitor supports automation tasks including:
 - Auto Welcome, Traceroute, **send LocalStats**
 - Remote admin scanner
- MeshMonitor stores all packet/node data locally in a SQLite DB.





Last 15 Minutes Telemetry

15m

1h

3h

6h

12h

24h

48h

3d

7d

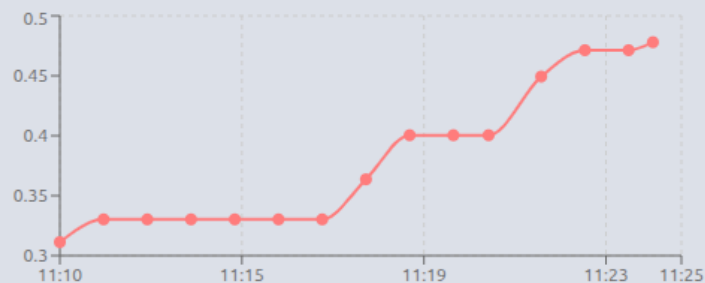
Active Nodes (MeshMonitor)

~ ☉ # ☆ ...



Air Utilization (TX) (%)

~ ☉ # ☆ ...



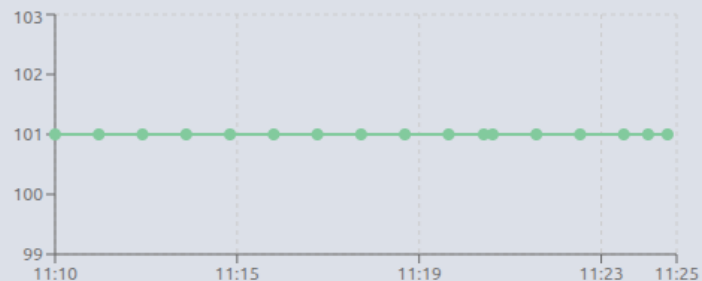
Bad Packets RX (Device) (packets)

~ ☉ # ☆ ...



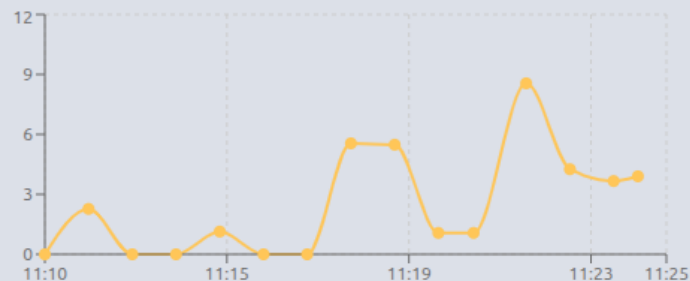
Battery Level (%)

~ ☉ # ☆ ...



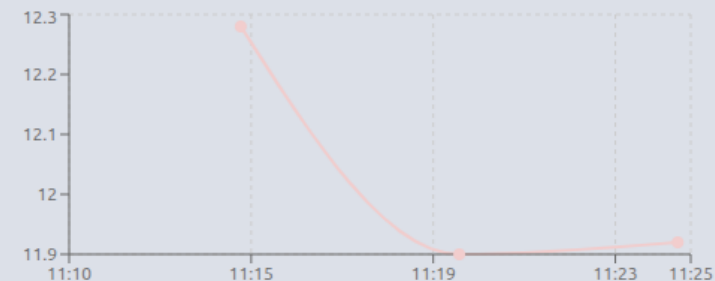
Channel Utilization (%)

~ ☉ # ☆ ...



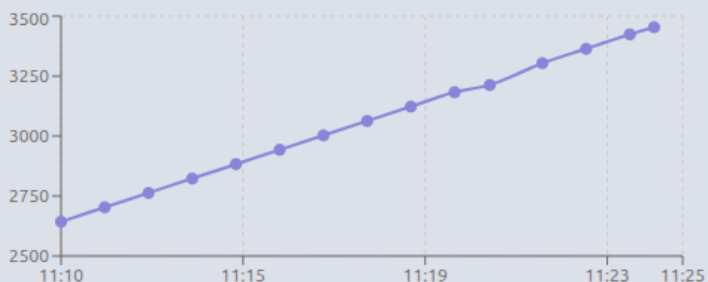
Clock Offset (Server - Node) (s)

~ ☉ # ☆ ...



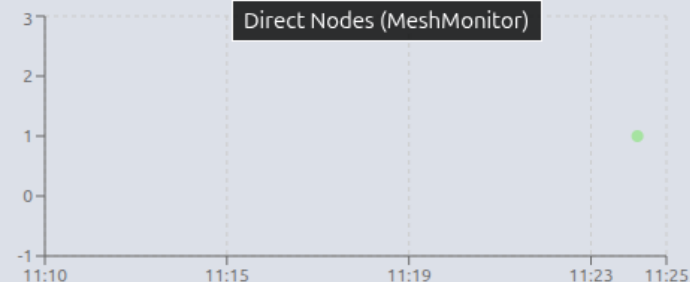
Device Uptime (s)

~ ☉ # ☆ ...



Direct Nodes (MeshMonitor)

~ ☉ # ☆ ...



Duplicate Packets (Device) (packets)

~ ☉ # ☆ ...



Telemetry Dashboard

Add widgets or star telemetry in the Nodes tab

+ Add Widgets

Days to View: 7

Search nodes or telemetry types...

All Nodes

All Types

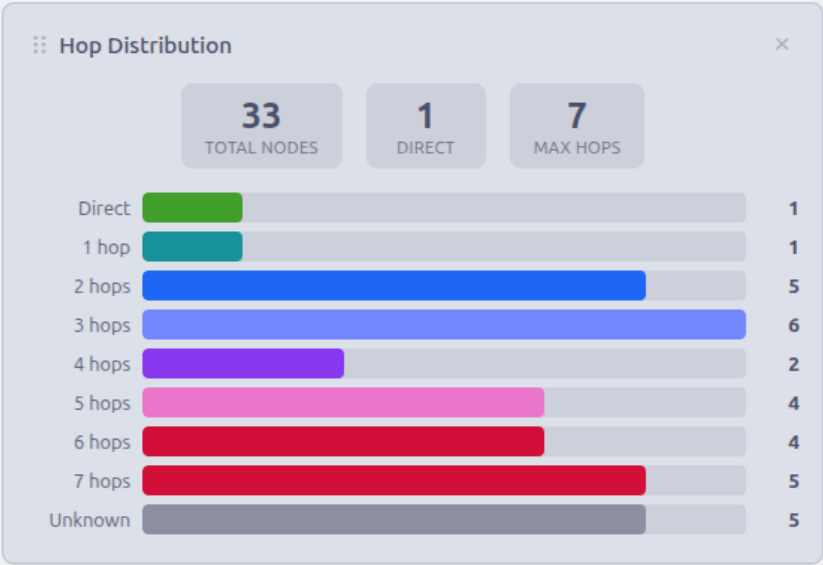
Device Roles: All

Sort by: Custom Order (Drag & Drop)

Node Status

Search to add a node...

Node	Last Heard	Signal/Hops	Voltage	Uptime	
Meshtastic 68b0	54s ago	-	4.34V	58m	×
Meshtastic b0cc	7m ago	1	-	-	×



Traceroute

Select a node...

Select a node above to view traceroute information.

Distance Distribution KM

Your node does not have GPS position data.

Save

Send Now

1

Nodes with Issues

0

Have Low-Entropy Keys

0

Have Duplicate Keys

0

Have Excessive Packets

1

Have Time Offset

Top Broadcasters (Last Hour)

Nodes with the highest packet counts in the last hour, regardless of spam status.

RANK	NODE	NODE ID	PACKETS/HOUR
#1	Meshtastic daa0 (daa0)	!6c73daa0	7
#2	Meshtastic b0cc (b0cc)	!dadfb0cc	6
#3	Meshtastic b8d4 (b8d4)	!dadfb8d4	6
#4	Meshtastic 68b0 (68b0)	!ea2468b0	1

Time Offset (1 nodes)

Nodes with clocks significantly out of sync with the server. This may indicate misconfigured devices or GPS issues.

Unknown (Unknown)

Node #F9A0762F - N/A *Last seen 1 days ago*

Time Offset

 1 day, 1 hour ahead ▶

Dead Nodes (1)

Select All

	NAME	ID	HARDWARE	LAST HEARD	LOCATION
☐	K1LRGR4M_TDECK (K1TD)	!9e019ae4	T_DECK	Never	 Local + Device

Target Node

Radio Configuration

Device Configuration

Module Configuration

Configuration Import/Export

Node Favorites & Ignored

Automatic Favorites

Target Node

Select the node to send admin commands to

Choose a local or remote node. Remote nodes will automatically request a session passkey.

Meshtastic 68b0

✓ Local node - no session passkey required

Load All Config

Retrieve Device Metadata

Send Reboot

Set Time

All local device configuration, including many features not available in Remote Admin, is available on the [Device Configuration](#) page.

▼ Radio Configuration

▶ LoRa Radio Configuration

Load

▶ Security Configuration

Load

▶ Channel Configuration

Load

▶ Device Configuration